Annotation Protocol of **RiceLeafDiseaseBD:** A high-resolution annotated smartphone image dataset of healthy and diseased rice leaves from Bangladesh

Dataset Characteristics

Table 1: Dataset Statistics Overview

Statistic	**Value**	**Details**
Total Images	9,769	6 classes
Total Bounding Boxes	21,460	Region-level annotations
Average Boxes per Image	$\sim 2.6^{\dagger}$	Varies by disease class
Image Resolution	1024×1024	Standardized preprocessing

† Averaged over the 8,194 images belonging to the five disease classes; healthy images carry empty annotation files.

1 Purpose

This document describes the standardized protocol developed for creating region-level annotations of diseased regions on rice leaves (*Oryza sativa*). The protocol aims to minimize subjectivity and variability in the annotation task, ensuring consistent and reproducible results across all annotators.

Protocol Objectives:
1. Standardized disease identification
2. Consistent bounding box placement
3. Expert-validated annotations
4. YOLO-compatible format output

Figure 1: Primary objectives of the annotation protocol

2 Scope

Region-level annotations are provided for images belonging to the following five disease classes:

Table 2: Disease Classes and Causal Agents

Disease Class	Causal Agent/Description
Blast	Caused by *Magnaporthe oryzae*
Brown Spot	Caused by *Bipolaris oryzae*
Leaf Smut	Caused by *Entyloma oryzae*
Rice Tungro	Viral disease transmitted by leafhoppers
Sheath Blight	Caused by *Rhizoctonia solani*
Healthy	No visible disease symptoms (empty annotations)

3 Annotation Strategy

Manual annotation challenges and solutions:

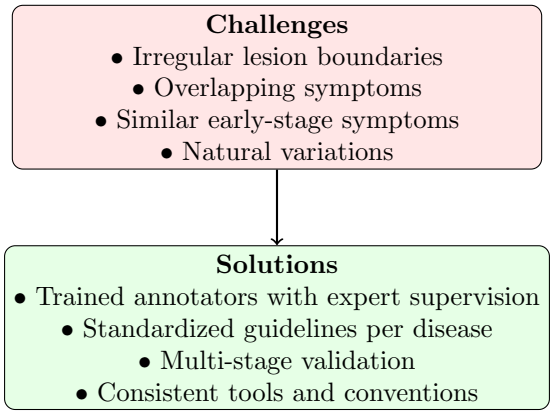


Figure 2: Annotation challenges and corresponding solutions

6 Annotation Guidelines by Disease Class

Table 3: Disease-Specific Annotation Guidelines

Disease	Annotation Rules
Blast	• Spindle-shaped lesions with gray-white centers • Brown margins • Individual lesions marked separately • Size: 1–2 cm length
Brown Spot	• Circular to oval spots • Brown centers • Each distinct spot individually • High density possible
Leaf Smut	• Small black spots/streaks • Follow leaf veins • All visible pustules included • Size: 2–5 mm
Rice Tungro	• Yellowing/discoloration patterns • Stunted growth areas • Mottled chlorotic patterns • Diffuse symptoms
Sheath Blight	• Large, irregular lesions • Greenish-gray to brown • Water-soaked borders • Size: 1–3 cm or larger

9 Conversion to YOLO Format

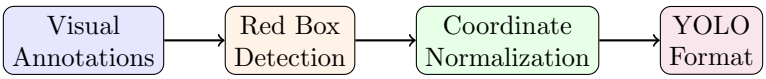


Figure 3: Conversion pipeline from visual annotations to YOLO format

9.2 Automated Extraction Procedure

Bounding-box coordinates are recovered from the drawn rectangles by colour segmentation followed by geometric validation:

1. **Stroke segmentation.** The image is converted to HSV and thresholded across both red hue lobes ($H \in [0, 6] \cup [174, 180]$) with a high saturation floor ($S \geq 150$, $V \geq 80$). The saturation constraint is essential: it separates the drawn stroke from naturally reddish content such as necrotic tissue, soil, dry vegetation and skin, which would otherwise be detected as annotations.

2. **Denoising.** A 2×2 morphological opening removes isolated speckles.

3. **Outline closure.** A morphological operation with a 15×15 elliptical kernel bridges small breaks in hand-drawn outlines so that each rectangle forms a single connected component.

4. **Component extraction.** Connected components with at least 25 stroke pixels are retained, and the axis-aligned bounding box of each is computed from its stroke pixels.

5. **Normalisation.** Coordinates are converted to YOLO format (class identifier, normalised centre, normalised width and height).

Images for which the procedure returns no boxes are flagged and annotated manually; 45 images (0.5%) required this step.

9.3 Annotation Revision History

An audit conducted during preparation of this release identified two defects in the initial coordinate conversion.

First, the reported annotation totals were inconsistent with the released label files. The published summary reported 74,836 boxes, whereas the label files contained 70,067.

Second, and more substantively, visual inspection of label boxes overlaid on their source images showed that the initial conversion detected reddish image content in addition to the drawn rectangles. Because the original procedure keyed on colour without geometric constraints, background regions such as soil, dry vegetation and, in some frames, the annotator's hand were recorded as lesion boxes. The drawn rectangles themselves were recovered correctly; the error was one of over-detection.

The manually drawn annotations are preserved in the visual overlay images and are unaffected by this issue. Coordinates have therefore been re-derived from those images using the procedure in Section 9.2, yielding **22,123** boxes across the five disease classes. Users of earlier versions of this dataset should migrate to the current release.

9.4 Class ID Mapping

Table 4: YOLO Format Class ID Mapping

Class ID	Disease Class	Sample Count	Avg. Boxes/Image	Total Boxes
0	Brown Spot	2,178	4.3	9,352
1	Blast	1,326	3.7	4,844
2	Healthy	1,575	–	–
3	Leaf Smut	724	2.1	1,501
4	Rice Tungro	2,244	1.4	3,156
5	Sheath Blight	1,722	1.5	2,607
Total		**9,769**	**2.6**	**21,460**

Listing 1: YOLO Format Example

```
# Example annotation file: brown_spot_001.txt
0 0.512695 0.651855 0.005859 0.006836
0 0.602051 0.639648 0.004883 0.005859
0 0.473145 0.839355 0.051758 0.083008

# Format: <class_id> <center_x> <center_y> <width> <height>
# All coordinates normalized to [0, 1] range
```

11 Quality Control Pipeline

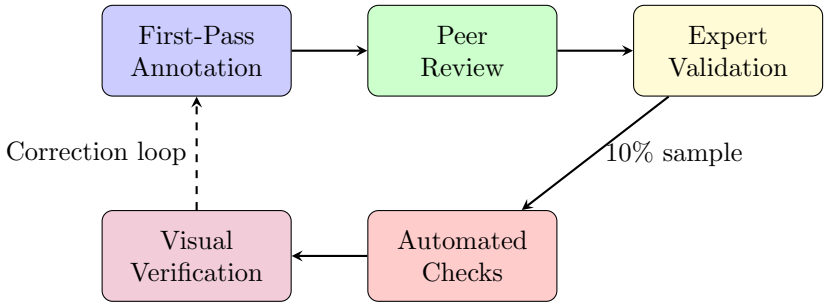


Figure 4: Quality control workflow with multi-stage validation

12 Inter-Annotator Agreement Results

Inter-Annotator Agreement Metrics (n=200 images)

Metric	Value	Interpretation	Visualization
Cohen's κ	0.82	Substantial agreement	82%
IoU Score	0.78	Good overlap	78%
F1-Score	0.85	High detection consistency	85%

Table 5: Inter-annotator agreement assessment results

15 Intended Use Cases

Table 6: Potential Applications and Use Cases

Application Area	Specific Use Cases
Computer Vision Research	• Benchmarking (YOLO, Faster R-CNN) • Transfer learning studies • Multi-task learning
Agricultural Applications	• Mobile diagnostic apps • Precision agriculture • Disease severity assessment
Data Science	• Lesion counting algorithms • Pattern recognition • Anomaly detection

16 Reproducibility Framework

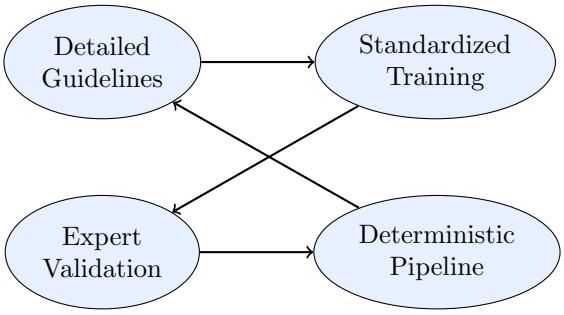


Figure 5: Reproducibility framework ensuring consistent annotations

17 Ethical Considerations

Ethical Framework

- Farm owner permissions obtained
- Respect for local agricultural practices
- Farmer expertise acknowledged and valued
- Dataset publicly available for agricultural research
- Commitment to ethical AI in agriculture

End of Annotation Protocol